```
#importing the packages
import pandas as pd
import numpy as np
cancer = pd.read_csv("/content/drive/MyDrive/ovariantotal.csv")
cancer.head()
```

{"type":"dataframe","variable_name":"cancer"}

```
cancer.shape
```

```
(349, 50)
```

```
X=cancer.drop('TYPE',axis=1)
y=cancer['TYPE']
```

fit_transform(X)

fit(X): Calculates the minimum and maximum values for each feature (i.e., for each column in X).

transform(X): Applies the following formula to each value in X:

$$X_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

```
from sklearn import preprocessing
min_max_scaler = preprocessing.MinMaxScaler()
X_scale = min_max_scaler.fit_transform(X)
print(X_scale)
```

```
[[2.45578349e-03 4.85071876e-01 4.70588235e-01 ... 8.37988827e-02
  7.60667904e-01 5.60447761e-01]
 [2.78074070e-02 6.55363067e-01 6.76470588e-01 ... 1.20111732e-01
  7.25417440e-01 4.32835821e-02]
 [7.35908185e-04 4.49686694e-01 3.52941176e-01 ... 3.43575419e-01
  8.34879406e-01 2.11194030e-01]
 ...
 [1.83563615e-03 4.80280133e-01 6.47058824e-01 ... 3.15642458e-01
  6.58627087e-01 2.11753731e-01]
 [1.01704165e-03 7.74788058e-01 2.20588235e-01 ... 2.45810056e-01
  7.99628942e-01 3.53917910e-01]
 [8.26863129e-04 2.50645042e-01 3.52941176e-01 ... 1.62011173e-01
  6.62337662e-01 1.65858209e-01]]
```

```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X_scale,y,test_size=0.2
0,random_state=0)
```

```
from tensorflow import keras
from keras.models import Sequential
from keras.layers import Dense
```

```
model = Sequential()
model.add(Dense(32, activation='relu', input_dim=49))
model.add(Dense(32, activation='relu'))
model.add(Dense(1, activation='sigmoid'))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an
`Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)
```

Given:

Dataset size = 349 samples

test_size=0.2 → 20% for testing, so Training set = 80% of 349 = 279 samples (approx.)

batch_size = 32

$$Steps\ per\ epoch = \left\lceil \frac{Number\ of\ training\ samples}{batch\ size} \right\rceil$$

```
model.compile(optimizer='adam',loss='binary_crossentropy',metrics=['ac
curacy'])
model.fit(X_train, y_train,batch_size=32, epochs=15)
```

```
Epoch 1/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.5134 - loss: 0.7014
Epoch 2/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6028 - loss: 0.6716

Epoch 3/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7619 - loss: 0.6497

Epoch 4/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7577 - loss: 0.6347

Epoch 5/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7290 - loss: 0.6180

Epoch 6/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7525 - loss: 0.6020

Epoch 7/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7548 - loss: 0.5870

Epoch 8/15
```

```
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7392 - loss: 0.5742

Epoch 9/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7744 - loss: 0.5395

Epoch 10/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7695 - loss: 0.5309

Epoch 11/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7878 - loss: 0.4890

Epoch 12/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7799 - loss: 0.4856

Epoch 13/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.7743 - loss: 0.5052

Epoch 14/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.7756 - loss: 0.4830

Epoch 15/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7924 - loss: 0.4500

<keras.src.callbacks.history.History at 0x7c6391f3a690>
```

```
y_pred=model.predict(X_test)
y_pred
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step

array([[0.6956166 ],
       [0.2946664 ],
       [0.69234145],
       [0.10522409],
       [0.8051121 ],
       [0.8608658 ],
       [0.87522113],
       [0.21871091],
       [0.10847906],
       [0.7466163 ],
       [0.7334747 ],
       [0.28958866],
       [0.83270323],
       [0.46649656],
       [0.11138376],
       [0.5524394 ],
       [0.6089235 ],
       [0.78218627],
       [0.89952147],
       [0.498813  ],
       [0.8099526 ],
```

```
       [0.8140079 ],
       [0.41399473],
       [0.82286954],
       [0.11483316],
       [0.76054394],
       [0.08680779],
       [0.2089046 ],
       [0.11289124],
       [0.7972676 ],
       [0.5098895 ],
       [0.36557412],
       [0.09772099],
       [0.21411562],
       [0.18500178],
       [0.03532215],
       [0.5627354 ],
       [0.7190877 ],
       [0.23296446],
       [0.32381338],
       [0.84445804],
       [0.77693623],
       [0.74228024],
       [0.7901234 ],
       [0.8343561 ],
       [0.06177657],
       [0.35265154],
       [0.74238074],
       [0.11735077],
       [0.34520686],
       [0.7503025 ],
       [0.73141974],
       [0.40398556],
       [0.39103752],
       [0.5101809 ],
       [0.7814058 ],
       [0.38098377],
       [0.82878405],
       [0.28687745],
       [0.8117017 ],
       [0.8426273 ],
       [0.30085582],
       [0.28724602],
       [0.83425325],
       [0.07550846],
       [0.5467596 ],
       [0.21025611],
       [0.6068192 ],
       [0.52263683],
       [0.25978714]], dtype=float32)
```

```python
y_pred = np.where(y_pred > 0.5, 1, 0)
y_pred = y_pred.astype(int)
import numpy as np
np.column_stack((y_pred, y_test))
```

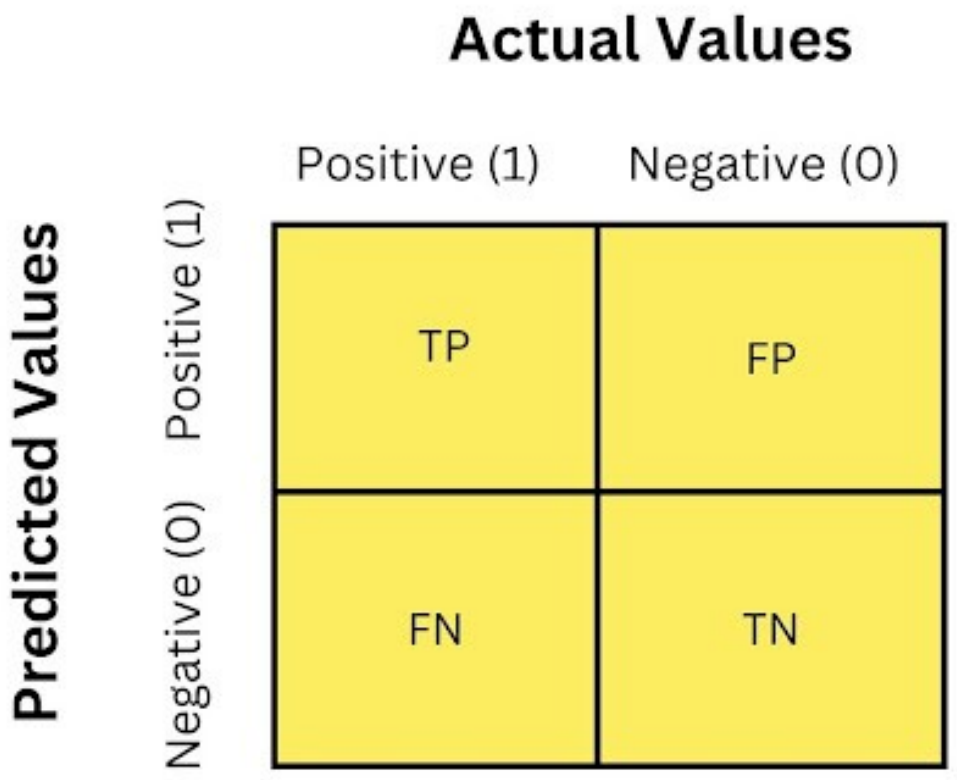
```
array([[1, 0],
       [0, 0],
       [1, 1],
       [0, 0],
       [1, 1],
       [1, 1],
       [1, 1],
       [0, 0],
       [0, 0],
       [1, 1],
       [1, 1],
       [0, 0],
       [1, 1],
       [0, 0],
       [0, 0],
       [1, 0],
       [1, 0],
       [1, 1],
       [1, 1],
       [0, 1],
       [1, 1],
       [1, 1],
       [0, 0],
       [1, 0],
       [0, 0],
       [1, 0],
       [0, 0],
       [0, 0],
       [0, 0],
       [1, 1],
       [1, 0],
       [0, 0],
       [0, 0],
       [0, 0],
       [0, 0],
       [0, 0],
       [1, 1],
       [1, 1],
       [0, 0],
       [0, 0],
       [1, 1],
       [1, 1],
       [1, 1],
       [1, 1],
       [1, 1],
```

```
       [0, 0],
       [0, 0],
       [1, 1],
       [0, 0],
       [0, 0],
       [1, 0],
       [1, 1],
       [0, 1],
       [0, 0],
       [1, 0],
       [1, 1],
       [0, 0],
       [1, 1],
       [0, 0],
       [1, 0],
       [1, 1],
       [0, 1],
       [0, 0],
       [1, 1],
       [0, 0],
       [1, 0],
       [0, 0],
       [1, 0],
       [1, 1],
       [0, 0]])
```

```
from sklearn.metrics import
confusion_matrix,classification_report,accuracy_score
print(confusion_matrix(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
```

```
[[30 11]
 [ 3 26]]
0.8
```

Confusion matrix



True Positive (TP): Model predicted 1 and actual is 1

False Positive (FP): Model predicted 1, but actual is 0 (Type I error)

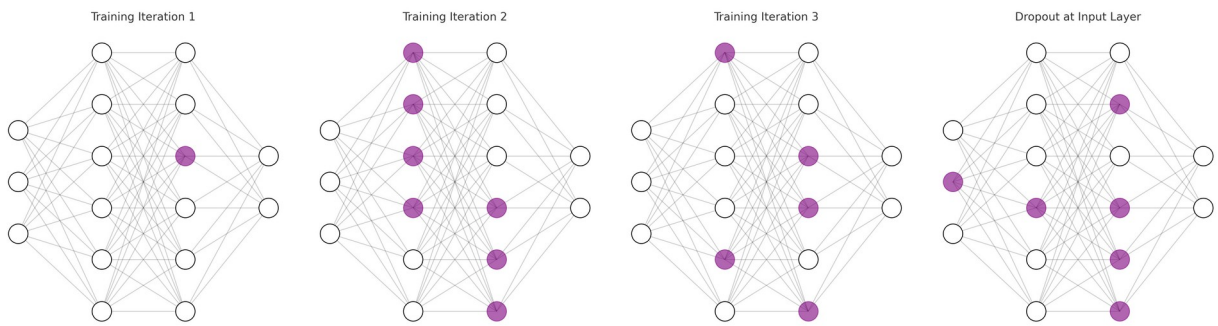
False Negative (FN): Model predicted 0, but actual is 1 (Type II error)

True Negative (TN): Model predicted 0 and actual is 0

Regularization Techniques

1. Dropout
2. Early stopping
3. L1 and L2 regularization
4. Batch normalization

1. Dropout




```
from tensorflow.keras.layers import Dropout
model = Sequential()

#model.add(Dropout(0.2, input_dim=49))
#model.add(Dense(32, activation='relu'))

model.add(Dense(32, activation='relu', input_dim=49))
model.add(Dropout(0.2))
model.add(Dense(32, activation='relu'))
model.add(Dropout(0.2))

#Output layer
model.add(Dense(1, activation='sigmoid'))


model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
model.fit(X_train, y_train,batch_size=32, epochs=15)
```

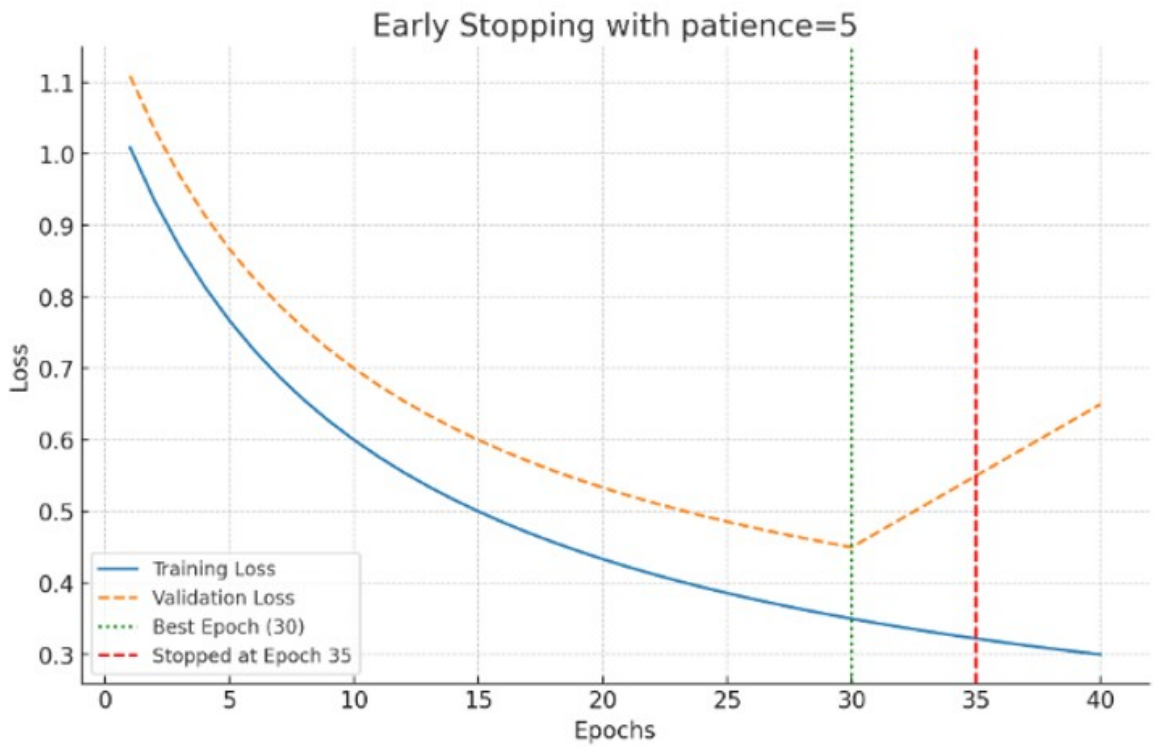
```
Epoch 1/15

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an
`Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

9/9 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.5135 - loss: 0.7016
Epoch 2/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.5586 - loss: 0.6817

Epoch 3/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.5721 - loss: 0.6735

Epoch 4/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6722 - loss: 0.6515

Epoch 5/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6549 - loss: 0.6271
```

```
Epoch 6/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6977 - loss: 0.6107

Epoch 7/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7194 - loss: 0.6057

Epoch 8/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6986 - loss: 0.6015

Epoch 9/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.6956 - loss: 0.5901

Epoch 10/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7494 - loss: 0.5651

Epoch 11/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7648 - loss: 0.5312

Epoch 12/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7905 - loss: 0.5130

Epoch 13/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7174 - loss: 0.5511

Epoch 14/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7770 - loss: 0.5025

Epoch 15/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.7707 - loss: 0.5147

<keras.src.callbacks.history.History at 0x7c639040d640>
```

```
y_pred=model.predict(X_test)
y_pred = np.where(y_pred > 0.5, 1, 0)
y_pred = y_pred.astype(int)
from sklearn.metrics import confusion_matrix,accuracy_score
print(confusion_matrix(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step
[[30 11]
 [ 3 26]]
0.8
```

Early stopping




```
from keras.callbacks import EarlyStopping
model = Sequential()
model.add(Dense(32, activation='relu', input_dim=49))
model.add(Dropout(0.2))
model.add(Dense(32, activation='relu'))
model.add(Dropout(0.2))
model.add(Dense(1, activation='sigmoid'))

model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
#watches the validation-loss metric during training.
# Early stopping
early_stop = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True)

history = model.fit(
    X_train, y_train,
    batch_size=32,
    epochs=100,
    validation_split=0.2, #20 % of X_train is set aside internally for
validation (so effectively 64 % train + 16 % validation + 20 % test).
    callbacks=[early_stop],
    verbose=1
)

# Evaluate performance
```

```
y_pred=model.predict(X_test)
y_pred = np.where(y_pred > 0.5, 1, 0)
y_pred = y_pred.astype(int)
from sklearn.metrics import confusion_matrix,accuracy_score
print(confusion_matrix(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
```

```
Epoch 1/100

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an
`Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

7/7 ━━━━━━━━━━━━━━━━━━━━ 3s 74ms/step - accuracy: 0.4824 - loss:
0.7097 - val_accuracy: 0.5714 - val_loss: 0.6887
Epoch 2/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - accuracy: 0.5662 - loss:
0.6856 - val_accuracy: 0.6607 - val_loss: 0.6739
Epoch 3/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - accuracy: 0.6669 - loss:
0.6580 - val_accuracy: 0.6786 - val_loss: 0.6560
Epoch 4/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - accuracy: 0.6392 - loss:
0.6445 - val_accuracy: 0.7143 - val_loss: 0.6378
Epoch 5/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - accuracy: 0.6783 - loss:
0.6281 - val_accuracy: 0.7143 - val_loss: 0.6235
Epoch 6/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.6536 - loss:
0.6240 - val_accuracy: 0.6964 - val_loss: 0.6133
Epoch 7/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.6903 - loss:
0.5961 - val_accuracy: 0.7143 - val_loss: 0.5978
Epoch 8/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.7562 - loss:
0.5696 - val_accuracy: 0.7143 - val_loss: 0.5881
Epoch 9/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.7392 - loss:
0.5401 - val_accuracy: 0.7679 - val_loss: 0.5808
Epoch 10/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.7296 - loss:
0.5366 - val_accuracy: 0.7500 - val_loss: 0.5643
Epoch 11/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.7626 - loss:
0.5299 - val_accuracy: 0.7679 - val_loss: 0.5625
Epoch 12/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7330 - loss:
```

```
0.5624 - val_accuracy: 0.7857 - val_loss: 0.5574
Epoch 13/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.7621 - loss:
0.5293 - val_accuracy: 0.7321 - val_loss: 0.5359
Epoch 14/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.7255 - loss:
0.5437 - val_accuracy: 0.7500 - val_loss: 0.5269
Epoch 15/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8077 - loss:
0.4899 - val_accuracy: 0.7857 - val_loss: 0.5262
Epoch 16/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.7961 - loss:
0.4807 - val_accuracy: 0.7857 - val_loss: 0.5355
Epoch 17/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7678 - loss:
0.4776 - val_accuracy: 0.7857 - val_loss: 0.5164
Epoch 18/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.7555 - loss:
0.4897 - val_accuracy: 0.7500 - val_loss: 0.5022
Epoch 19/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7383 - loss:
0.5183 - val_accuracy: 0.8036 - val_loss: 0.5094
Epoch 20/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.7898 - loss:
0.4656 - val_accuracy: 0.7857 - val_loss: 0.5214
Epoch 21/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.7721 - loss:
0.4752 - val_accuracy: 0.8036 - val_loss: 0.4933
Epoch 22/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.7727 - loss:
0.4812 - val_accuracy: 0.8036 - val_loss: 0.4999
Epoch 23/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7742 - loss:
0.4549 - val_accuracy: 0.8036 - val_loss: 0.4921
Epoch 24/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.7865 - loss:
0.4599 - val_accuracy: 0.8036 - val_loss: 0.4906
Epoch 25/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.7962 - loss:
0.4262 - val_accuracy: 0.8036 - val_loss: 0.4991
Epoch 26/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8392 - loss:
0.3890 - val_accuracy: 0.8036 - val_loss: 0.4900
Epoch 27/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8540 - loss:
0.4044 - val_accuracy: 0.8036 - val_loss: 0.4853
Epoch 28/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8352 - loss:
0.3942 - val_accuracy: 0.8036 - val_loss: 0.4841
```

```
Epoch 29/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.7806 - loss:
0.4533 - val_accuracy: 0.7857 - val_loss: 0.5004
Epoch 30/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8096 - loss:
0.4227 - val_accuracy: 0.8036 - val_loss: 0.4934
Epoch 31/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8082 - loss:
0.4271 - val_accuracy: 0.8214 - val_loss: 0.4747
Epoch 32/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8126 - loss:
0.4058 - val_accuracy: 0.8036 - val_loss: 0.4803
Epoch 33/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8496 - loss:
0.3537 - val_accuracy: 0.8036 - val_loss: 0.4839
Epoch 34/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.7933 - loss:
0.4242 - val_accuracy: 0.8036 - val_loss: 0.4828
Epoch 35/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.8559 - loss:
0.3645 - val_accuracy: 0.8214 - val_loss: 0.4765
Epoch 36/100
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8598 - loss:
0.3827 - val_accuracy: 0.8036 - val_loss: 0.4878
```

```
WARNING:tensorflow:5 out of the last 7 calls to <function
TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_dist
ributed at 0x7c6379c7ec00> triggered tf.function retracing. Tracing is
expensive and the excessive number of tracings could be due to (1)
creating @tf.function repeatedly in a loop, (2) passing tensors with
different shapes, (3) passing Python objects instead of tensors. For
(1), please define your @tf.function outside of the loop. For (2),
@tf.function has reduce_retracing=True option that can avoid
unnecessary retracing. For (3), please refer to
https://www.tensorflow.org/guide/function#controlling_retracing and
https://www.tensorflow.org/api_docs/python/tf/function for  more
details.
```

```
 1/3 ━━━━━━━━━━━━━━━━━━━━ 0s 55ms/step
```

```
WARNING:tensorflow:6 out of the last 9 calls to <function
TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_dist
ributed at 0x7c6379c7ec00> triggered tf.function retracing. Tracing is
expensive and the excessive number of tracings could be due to (1)
creating @tf.function repeatedly in a loop, (2) passing tensors with
different shapes, (3) passing Python objects instead of tensors. For
(1), please define your @tf.function outside of the loop. For (2),
@tf.function has reduce_retracing=True option that can avoid
unnecessary retracing. For (3), please refer to
https://www.tensorflow.org/guide/function#controlling_retracing and
```

```
https://www.tensorflow.org/api_docs/python/tf/function for  more
details.

 3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
[[32  9]
 [ 2 27]]
0.8428571428571429
```

L1 and L2 regularization

$$L = \frac{1}{N}\sum_i L_i(f(x_i, W), y_i) + \lambda R(W)$$

```
from tensorflow.keras import layers, models, regularizers

model = Sequential()

model.add(Dense(32, activation='relu',
kernel_regularizer=regularizers.L1(0.001), input_dim=49))
model.add(Dense(32, kernel_regularizer=regularizers.L1(0.001),
activation='relu'))

#Output layer
model.add(Dense(1, activation='sigmoid'))


model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
model.fit(X_train, y_train,batch_size=32, epochs=15)

Epoch 1/15
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an
`Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

9/9 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.4518 - loss: 1.0679
Epoch 2/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.5722 - loss: 1.0370

Epoch 3/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6782 - loss: 1.0114

Epoch 4/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7647 - loss: 0.9763

Epoch 5/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7217 - loss: 0.9485

Epoch 6/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7463 - loss: 0.9045

Epoch 7/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7724 - loss: 0.8657

Epoch 8/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7851 - loss: 0.8459

Epoch 9/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7852 - loss: 0.8094

Epoch 10/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7946 - loss: 0.7869

Epoch 11/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7816 - loss: 0.7483

Epoch 12/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.8105 - loss: 0.7509

Epoch 13/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.8011 - loss: 0.7243

Epoch 14/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.8021 - loss: 0.7026

Epoch 15/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.8027 - loss: 0.6831

<keras.src.callbacks.history.History at 0x7c6379b95b80>
```

```
y_pred=model.predict(X_test)
y_pred = np.where(y_pred > 0.5, 1, 0)
y_pred = y_pred.astype(int)
from sklearn.metrics import
confusion_matrix,classification_report,accuracy_score
print(confusion_matrix(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
[[33  8]
 [ 3 26]]
0.8428571428571429
```

```
from tensorflow.keras import layers, models, regularizers

model = Sequential()
model.add(Dense(32, activation='relu',
kernel_regularizer=regularizers.L2(0.001), input_dim=49))
model.add(Dense(32, kernel_regularizer=regularizers.L2(0.001),
activation='relu'))

#Output layer
model.add(Dense(1, activation='sigmoid'))

model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
model.fit(X_train, y_train,batch_size=32, epochs=15)
```

```
Epoch 1/15

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an
`Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

9/9 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step - accuracy: 0.4823 - loss: 0.7735
Epoch 2/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6317 - loss: 0.7416

Epoch 3/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7121 - loss: 0.7190

Epoch 4/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.6990 - loss: 0.7025

Epoch 5/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.7148 - loss: 0.6839

Epoch 6/15
```

```
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7541 - loss: 0.6491

Epoch 7/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6976 - loss: 0.6446

Epoch 8/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7259 - loss: 0.6242

Epoch 9/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.7295 - loss:
0.6142
Epoch 10/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.7547 - loss:
0.5673
Epoch 11/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.7446 - loss: 0.5720

Epoch 12/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.7596 - loss: 0.5421

Epoch 13/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.7944 - loss: 0.5006

Epoch 14/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.7442 - loss: 0.5391

Epoch 15/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.7971 - loss:
0.4948
```

```
<keras.src.callbacks.history.History at 0x7c6379be4e60>
```

```
y_pred=model.predict(X_test)
y_pred = np.where(y_pred > 0.5, 1, 0)
y_pred = y_pred.astype(int)
from sklearn.metrics import
confusion_matrix,classification_report,accuracy_score
print(confusion_matrix(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step
[[30 11]
 [ 4 25]]
0.7857142857142857
```

```
from tensorflow import keras
from keras.models import Sequential
from keras.layers import Dense, BatchNormalization, Dropout

# Define the Sequential model
model = Sequential()
```

```python
# Input layer + 1st hidden layer
model.add(Dense(32, activation='relu', input_dim=49))
model.add(BatchNormalization())          # Normalize activations to
stabilize learning
model.add(Dropout(0.3))                  # Optional: Dropout for
regularization

# 2nd hidden layer
model.add(Dense(32, activation='relu'))
model.add(BatchNormalization())          # Normalize the second
layer's activations
model.add(Dropout(0.3))

# Output layer
model.add(Dense(1, activation='sigmoid'))  # Binary classification
output

# Compile the model
model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
model.fit(X_train, y_train,batch_size=32, epochs=15)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an
`Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

Epoch 1/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - accuracy: 0.4941 - loss: 0.9027
Epoch 2/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6658 - loss: 0.6681

Epoch 3/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.7011 - loss: 0.6034

Epoch 4/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.6251 - loss: 0.6813

Epoch 5/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7247 - loss: 0.5546

Epoch 6/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.7247 - loss: 0.5752

Epoch 7/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7630 - loss: 0.4875

Epoch 8/15
```

```
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7214 - loss: 0.6054

Epoch 9/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.7535 - loss: 0.4741

Epoch 10/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7667 - loss: 0.5297

Epoch 11/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7464 - loss: 0.5120

Epoch 12/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7885 - loss: 0.4761

Epoch 13/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7394 - loss: 0.5114

Epoch 14/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8007 - loss: 0.4958

Epoch 15/15
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.7774 - loss: 0.4697

<keras.src.callbacks.history.History at 0x7c637a2f8bc0>
```

```
y_pred=model.predict(X_test)
y_pred = np.where(y_pred > 0.5, 1, 0)
y_pred = y_pred.astype(int)
from sklearn.metrics import
confusion_matrix,classification_report,accuracy_score
print(confusion_matrix(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step
[[24 17]
 [ 1 28]]
0.7428571428571429
```